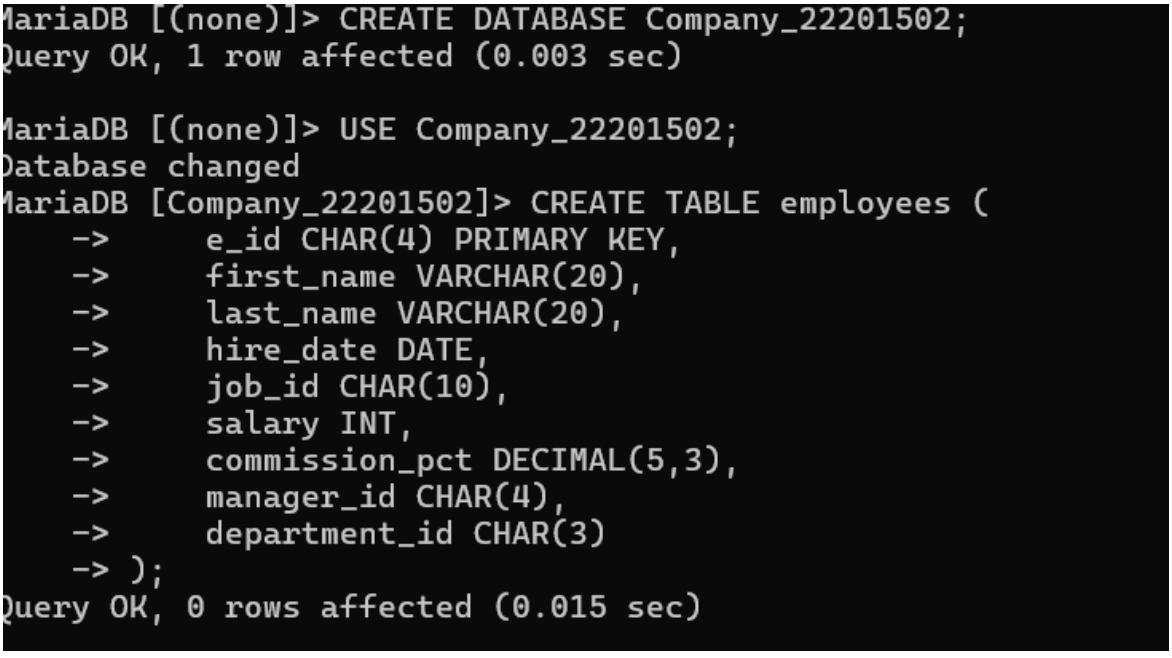
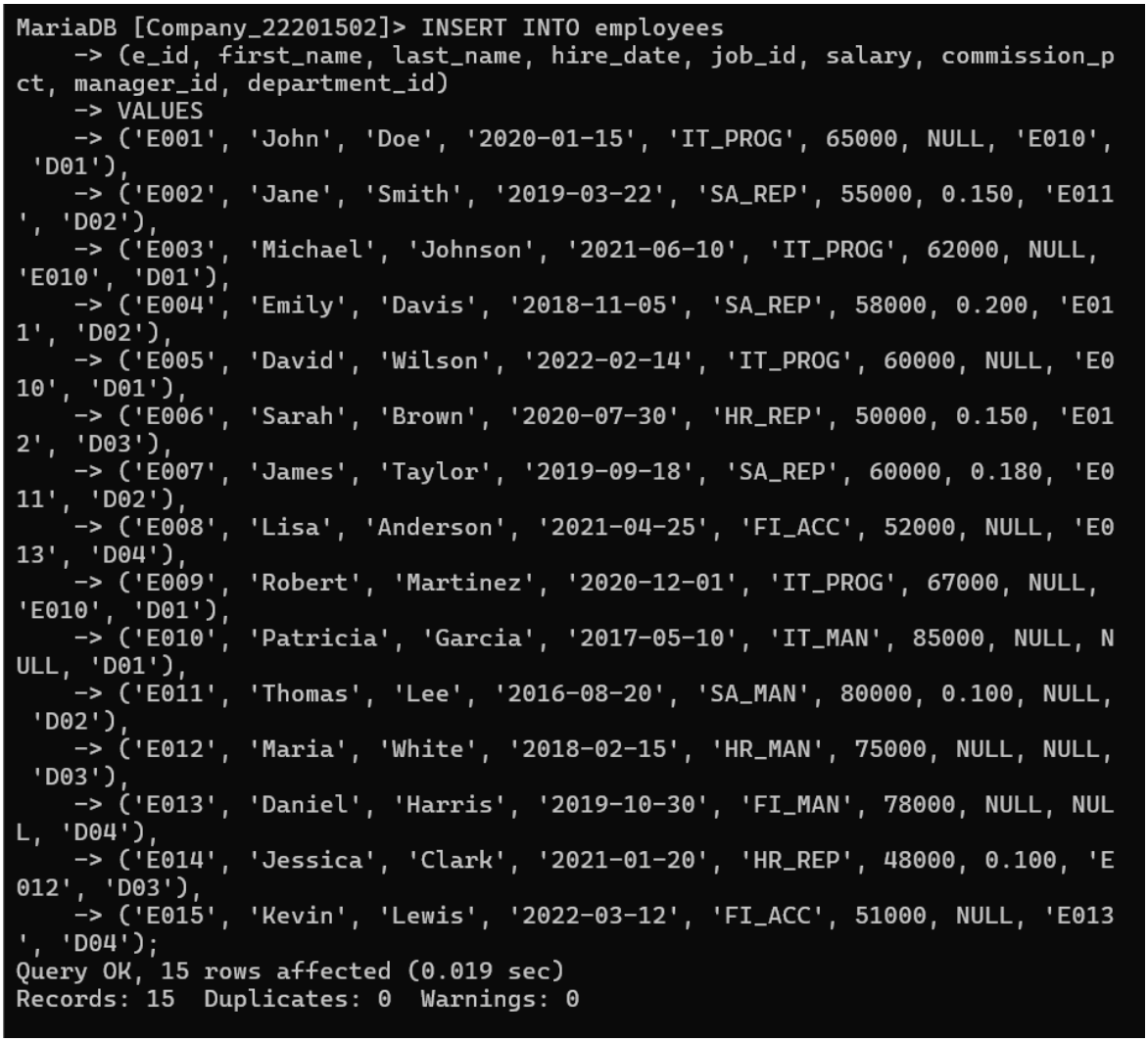
CSE370 : Database Systems

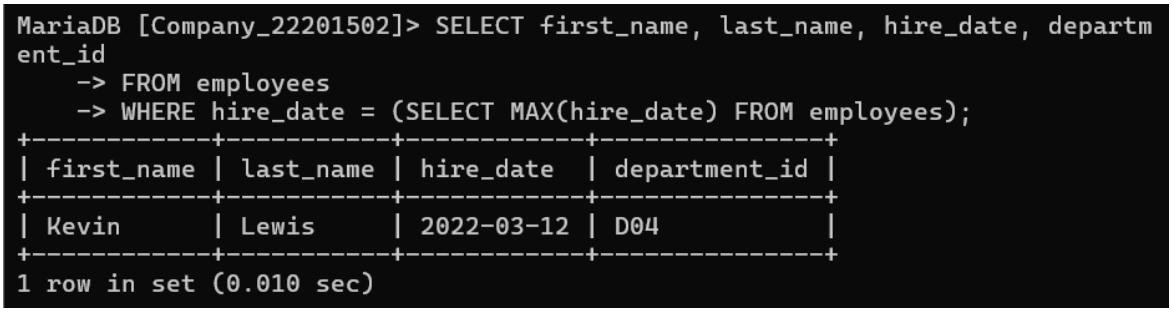
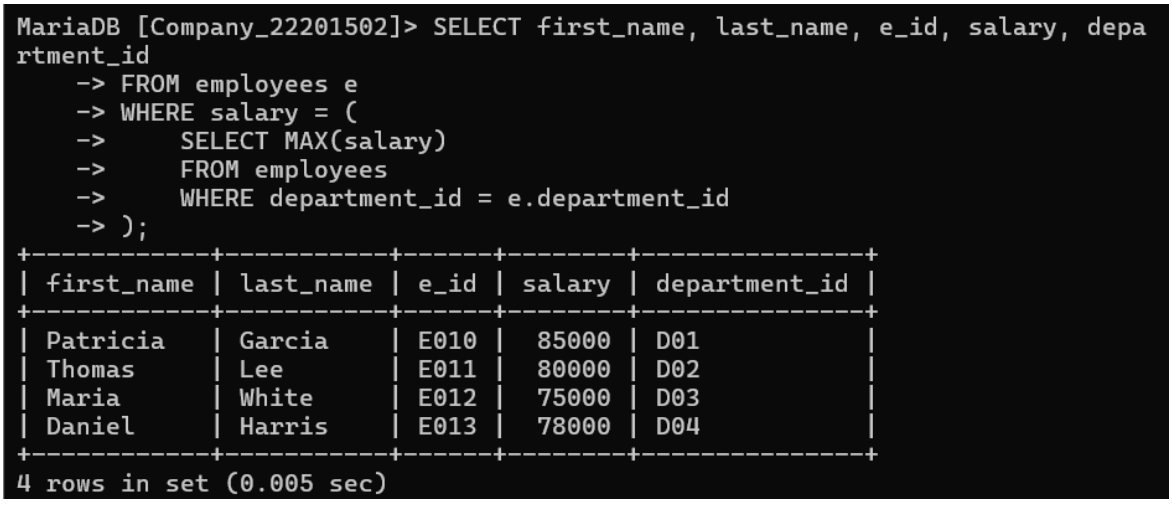
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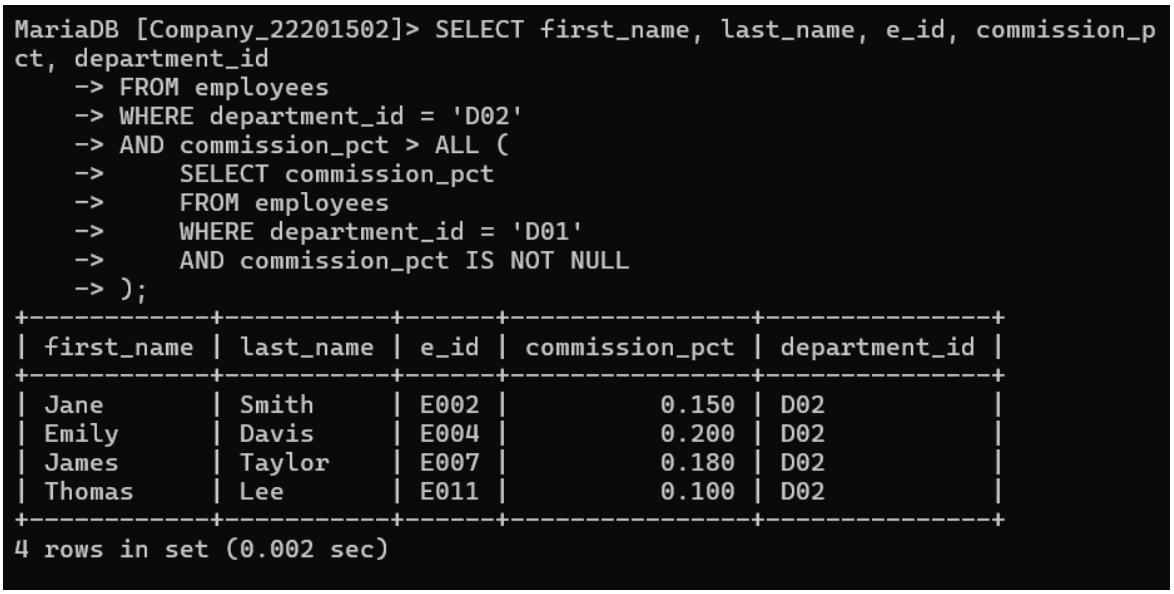
ID : 22201502 | Name : Abdullah al mahmud

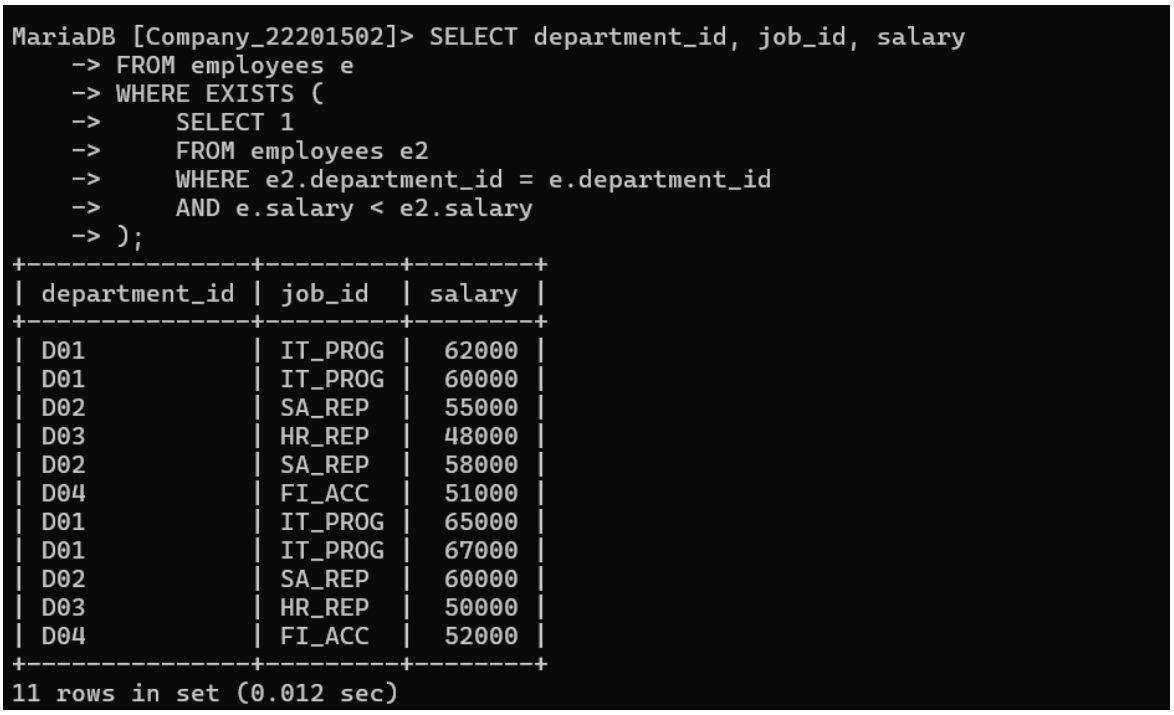
No 1 Query (as Plain Text)	CREATE TABLE employees (e_id CHAR(4) PRIMARY KEY, first_name VARCHAR(20), last_name VARCHAR(20), hire_date DATE, job_id CHAR(10), salary INT, commission_pct DECIMAL(5,3), manager_id CHAR(4), department_id CHAR(3));
No 1 SS (of Query & Output in Shell)	
No 2 Query (as Plain Text)	INSERT INTO employees (e_id, first_name, last_name, hire_date, job_id, salary, commission_pct, manager_id, department_id)

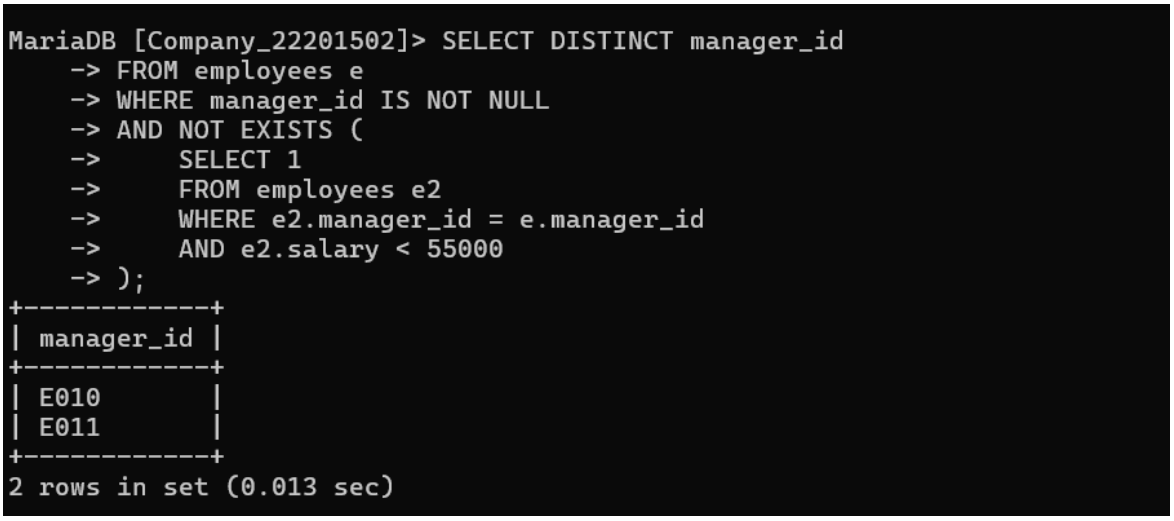
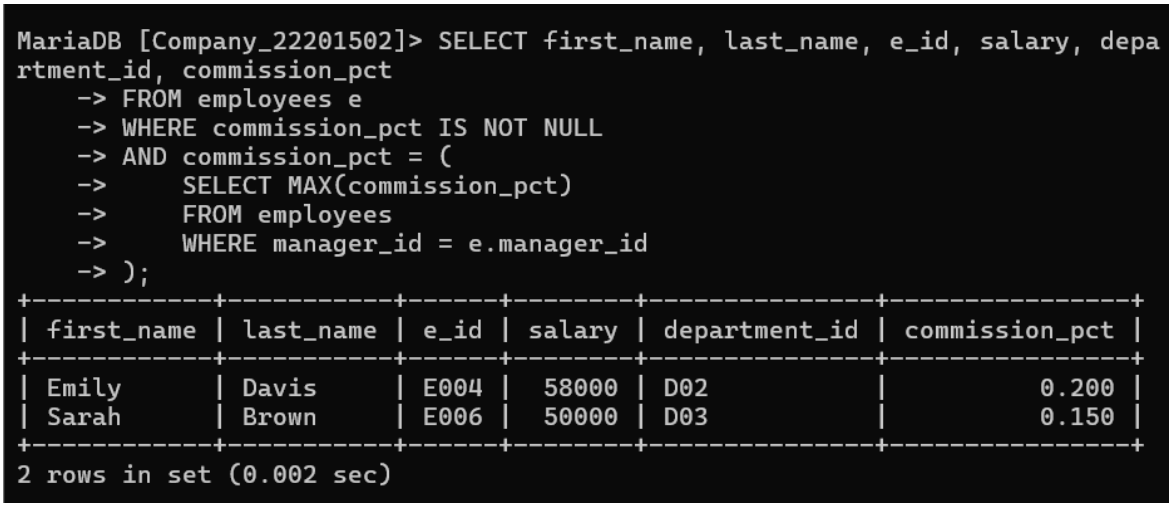
	VALUES ('E001', 'John', 'Doe', '2020-01-15', 'IT_PROG', 65000, NULL, 'E010', 'D01'), ('E002', 'Jane', 'Smith', '2019-03-22', 'SA_REP', 55000, 0.150, 'E011', 'D02'), ('E003', 'Michael', 'Johnson', '2021-06-10', 'IT_PROG', 62000, NULL, 'E010', 'D01'), ('E004', 'Emily', 'Davis', '2018-11-05', 'SA_REP', 58000, 0.200, 'E011', 'D02'), ('E005', 'David', 'Wilson', '2022-02-14', 'IT_PROG', 60000, NULL, 'E010', 'D01'), ('E006', 'Sarah', 'Brown', '2020-07-30', 'HR_REP', 50000, 0.150, 'E012', 'D03'), ('E007', 'James', 'Taylor', '2019-09-18', 'SA_REP', 60000, 0.180, 'E011', 'D02'), ('E008', 'Lisa', 'Anderson', '2021-04-25', 'FI_ACC', 52000, NULL, 'E013', 'D04'), ('E009', 'Robert', 'Martinez', '2020-12-01', 'IT_PROG', 67000, NULL, 'E010', 'D01'), ('E010', 'Patricia', 'Garcia', '2017-05-10', 'IT_MAN', 85000, NULL, NULL, 'D01'), ('E011', 'Thomas', 'Lee', '2016-08-20', 'SA_MAN', 80000, 0.100, NULL, 'D02'), ('E012', 'Maria', 'White', '2018-02-15', 'HR_MAN', 75000, NULL, NULL, 'D03'), ('E013', 'Daniel', 'Harris', '2019-10-30', 'FI_MAN', 78000, NULL, NULL, 'D04'), ('E014', 'Jessica', 'Clark', '2021-01-20', 'HR_REP', 48000, 0.100, 'E012', 'D03'), ('E015', 'Kevin', 'Lewis', '2022-03-12', 'FI_ACC', 51000, NULL, 'E013', 'D04');

No 2 SS (of Query & Output in Shell)	
No 3 Query (as Plain Text)	SELECT first_name, last_name, hire_date, department_id FROM employees WHERE hire_date = (SELECT MAX(hire_date) FROM employees);

No 3 SS (of Query & Output in Shell)	
No 4 Query (as Plain Text)	SELECT first_name, last_name, e_id, salary, department_id FROM employees e WHERE salary = (SELECT MAX(salary) FROM employees WHERE department_id = e.department_id);
No 4 SS (of Query & Output in Shell)	
No 5 Query (as Plain Text)	SELECT first_name, last_name, e_id, commission_pct, department_id FROM employees WHERE department_id = 'D02' AND commission_pct > ALL (SELECT commission_pct FROM employees WHERE department_id = 'D01'

	AND commission_pct IS NOT NULL);
No 5 SS (of Query & Output in Shell)	
No 6 Query (as Plain Text)	SELECT department_id, COUNT(*) AS total_employees FROM employees GROUP BY department_id HAVING MAX(salary) <= 70000;
No 6 SS (of Query & Output in Shell)	
No 7 Query (as Plain Text)	SELECT department_id, job_id, salary FROM employees e WHERE EXISTS (SELECT 1 FROM employees e2 WHERE e2.department_id = e.department_id AND e.salary < e2.salary

	);
No 7 SS (of Query & Output in Shell)	
No 8 Query (as Plain Text)	SELECT DISTINCT manager_id FROM employees e WHERE manager_id IS NOT NULL AND NOT EXISTS (SELECT 1 FROM employees e2 WHERE e2.manager_id = e.manager_id AND e2.salary < 55000);

No 8 SS (of Query & Output in Shell)	
No 9 Query (as Plain Text)	SELECT first_name, last_name, e_id, salary, department_id, commission_pct FROM employees e WHERE commission_pct IS NOT NULL AND commission_pct = (SELECT MAX(commission_pct) FROM employees WHERE manager_id = e.manager_id);
No 9 SS (of Query & Output in Shell)	
No 10 Query	SELECT job_id, AVG(salary) AS average_salary

(as Plain Text)	FROM employees GROUP BY job_id HAVING AVG(salary) < (SELECT AVG(salary) FROM employees);
No 10 SS (of Query & Output in Shell)	